# 1. Clinical Question & Core Hypothesis

## Clinical Question

Tiered analysis highlights that non-invasive signals (urine-DNA methylation, CTCs) and immune context (Immunoscore) do not uniformly predict favorable outcomes—some NMIBC tumors with apparent immune infiltration still progress. Why does local immune organization (TLS/TILs) fail to prevent progression in a subset of NMIBC? 2) Tier 1 paths used: I built on MSH6 (MMR-associated genomic instability) and urine-DNA methylation biomarkers because they provide complementary mechanistic axes—genomic alteration shaping neoantigen and immune-editing dynamics (MSH6) and stable epigenetic programs (urine methylation) that can silence antigen-presentation or immune-regulatory genes and are clinically measurable non-invasively. I deliberately used CTCs as a secondary modifier (systemic dissemination signal) rather than an anchor to keep the hypothesis focused on local immune–tumor epigenomic interactions. 3) Tier 2 support: TGF-β–driven EMT/fibrotic signaling can remodel stroma and suppress effective lymphoid organization; E-FABP–linked metabolic reprogramming can promote invasive phenotypes that resist immune control; Immunoscore/immune-density measures corroborate that immune quantity alone may be insufficient without functional antigen presentation. These support a mechanism where epigenetic silencing and microenvironmental immunosuppression negate TLS benefit. 4) Endpoint choice: Progression-free interval (PFI) is the most clinically relevant endpoint for NMIBC because the key clinical problem is progression to muscle-invasive disease (worse morbidity/mortality) and PFI directly captures that risk, making the hypothesis actionable for surveillance and early-treatment decisions. 5) Deliberate exclusions: I excluded CTCs and AI-based methods from the core mechanistic hypothesis to avoid conflating local immune–epigenetic mechanisms with systemic detection/implementation issues; GLU/renal markers and CSF/AD-related paths were omitted as they lack direct mechanistic linkage to NMIBC progression. This yields a sharp, testable question linking TLS functionality, MSH6 status, and urine-methylation–mediated antigen-presentation suppression?

## Working Hypothesis

In NMIBC, TLS-like lymphoid aggregates fail to confer durable anti-tumor control when tumors harbor MSH6 deficiency, because MSH6-driven mismatch repair loss increases somatic hypermutation and promotes immune editing while concomitant urine-DNA methylation signatures silence antigen-presentation pathways, thereby blunting TLS function. Therefore, TLS-rich NMIBC tumors with MSH6 loss and high urine-DNA methylation will have shorter progression-free intervals, identifying patients who would benefit from intensified surveillance or earlier escalation to intravesical or systemic therapies, with CTCs as a secondary modifier of risk.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | MSH6, urine-DNA methylation signature |
| Primary endpoint | PFI |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**MSH6:** MSH6 (MMR-associated genomic instability) and urine-DNA methylation biomarkers because they provide complementary mechanistic axes—genomic alteration shaping neoantigen and immune-editing dynamics (MSH6) and stable epigenetic programs (urine methylation) that can silence antigen-presentation or immune-regulatory genes and are clinically measurable non-invasively.

**urine-DNA methylation signature:** Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | CTCs (circulating tumor cells) | Supporting modifier from knowledge graph |
| Secondary | TGF-β signaling | Supporting modifier from knowledge graph |
| Secondary | E-FABP (fatty-acid binding protein) | Supporting modifier from knowledge graph |
| Secondary | Immunoscore | Supporting modifier from knowledge graph |
| Secondary | AI-based biomarker integration | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** cohort with NMIBC tissue to assess TLS status histologically;

**Prediction 2:** MSH6 status by IHC/genomic assays;

**Prediction 3:** urine DNA methylation profiling for antigen-presentation gene signatures;

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

### Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

### Alternative Directions

- CTCs (circulating tumor cells): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TGF-β signaling: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.